

The Internet of Things, And Beyond

The Internet of Things is by no means a new topic. However, the initial definition no longer holds true today as a sea change has happened in the last 30 years of the Internet of Things timeline. What started off with RFID tags simply identifying different devices has now grown into a system envisioned to automate almost every aspect of human existence. Read this article to understand how to capitalise on this opportunity



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Until now, the Internet of Things (IoT) system was limited to items with RFID tags that allowed automated monitoring and servicing of processes. The system allowed inventory of everything by computers—groceries, tech equipment, tools and even the humans that used them. The data made available to the system by this inventory, allowed the

system to make decisions like dumping of expired consumables or re-ordering of stock for goods that were almost exhausted. Overall, this system took care of most of the repetitive and time-consuming tasks smoothly.

The IoT has now evolved to encompass even more electronics, and as the cloud improves, the IoT has leveraged it to massively revamp its functionality. Smartphone—a humble device in everybody's hands today—is a cocktail of sensors that offer a significant amount of data to IoT applications—traffic predictions (Google Maps), healthcare (Nike Fuel+Band) and even home au-

tomation (Nullriver Haiku) are now easily possible.

There are numerous benefits that a more widespread transition to the

IoT could bring about. "This transition will essentially spawn new technologies, and change paradigms of how we do things and how businesses operate. In the long term, this will have a significant societal impact as well," explained Dr P.A. Subrahmanyam, executive chairman, CyberKnowledge, in his talk 'The Internet of Things—Opportunities and Challenges' at Electronics Rocks 2012 conference held recently in Bengaluru.

So how do you get about doing it?

Leverage RFID technology. Due to the popularity of RFID systems, RFID technology is thought to be a prerequisite for the Internet of Things. RFID tags have developed and come a long way since the initial release in the 1990s, when there were a collection of different standards that unfortunately acted to prohibit RFID's growth. That changed with the adoption of ISO 18000-6C in 2006, which is a second-generation air-interface protocol. ISO 18000-6C is a popular choice among IoT designers implementing RFID technology. Most readers in the market that target the Internet of Things are based on this standard.

RFID tags add value to a physical item in the IoT by creating a virtual object of that item and populating it

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